# Agenda

**Card Dispatch Process Automation**

1 Current Process (NTB)

2 Current Process (ETB)

3 Manual Interventions

4 Proposed High Level Process

5 Proposed Process (NTB)

6 Proposed Process (ETB)

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8 Automation Areas

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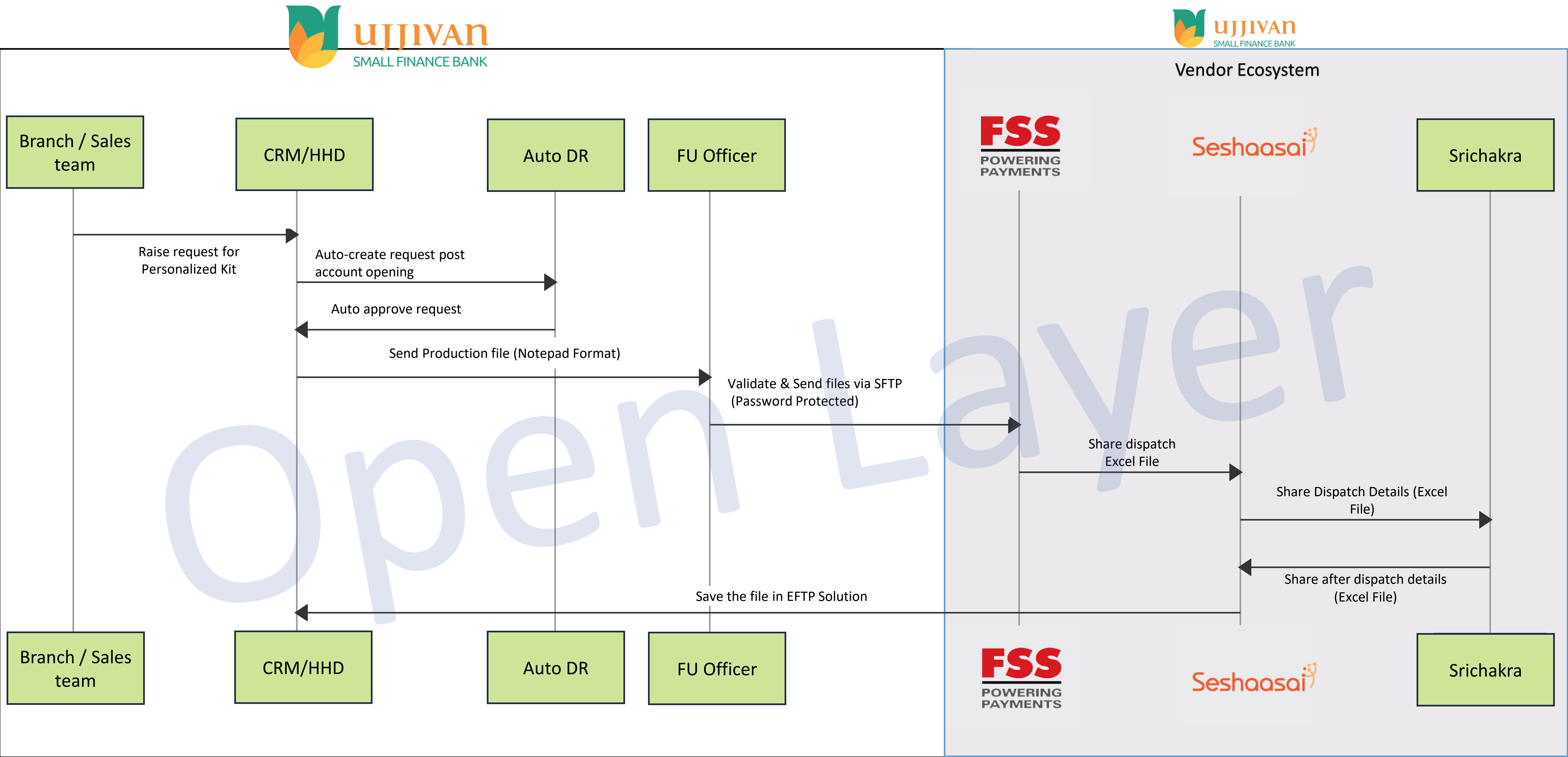
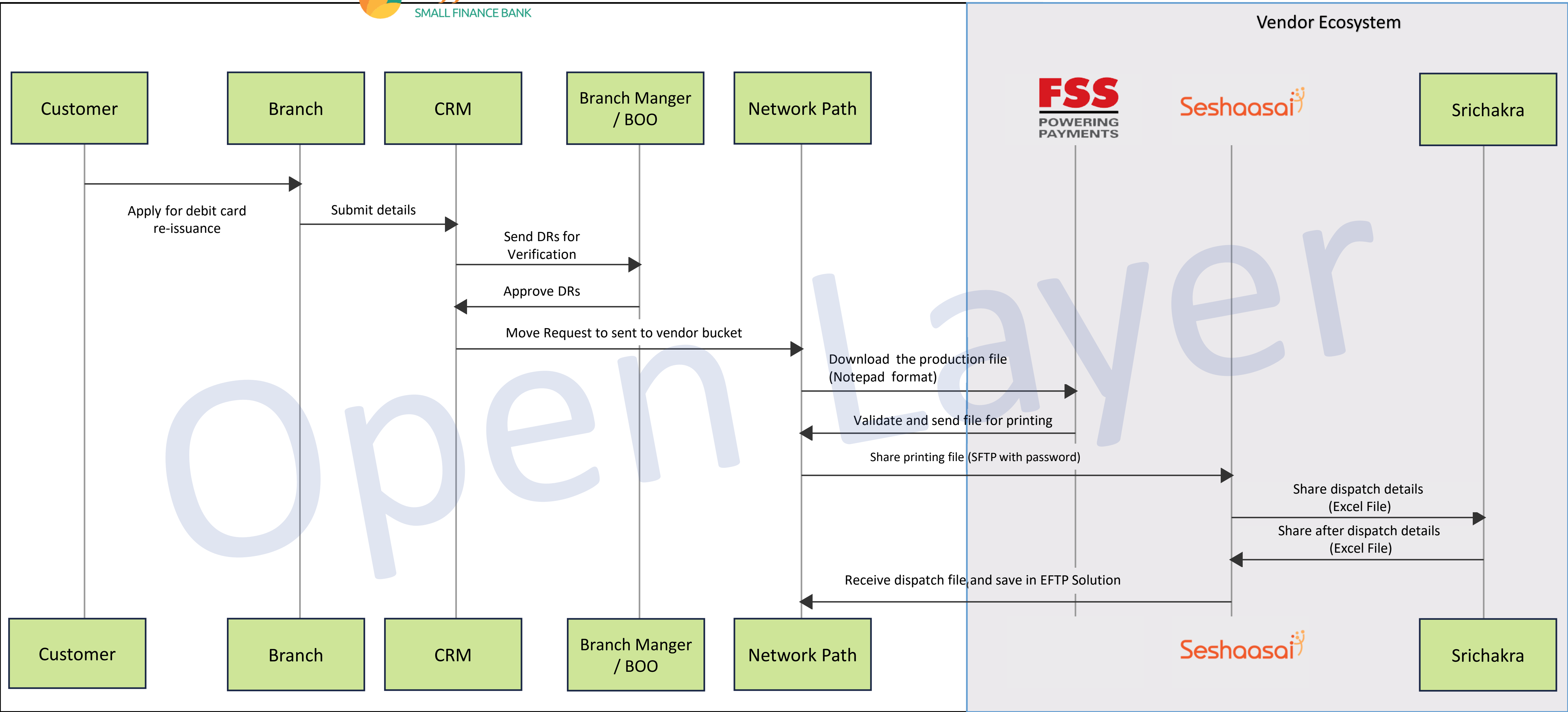
# Current Process for Cards(NTB)



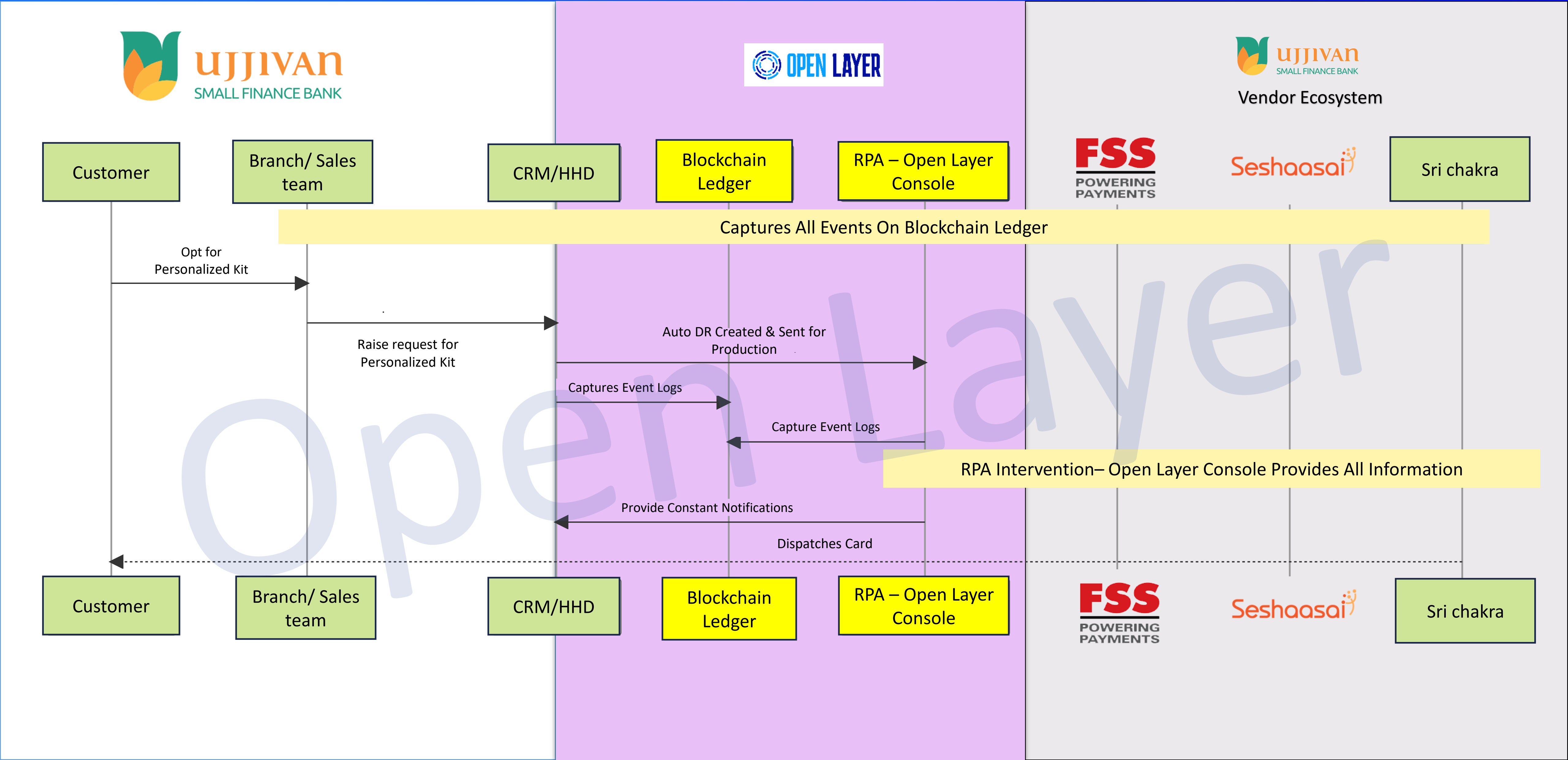
# Current Process for Cards(ETB)



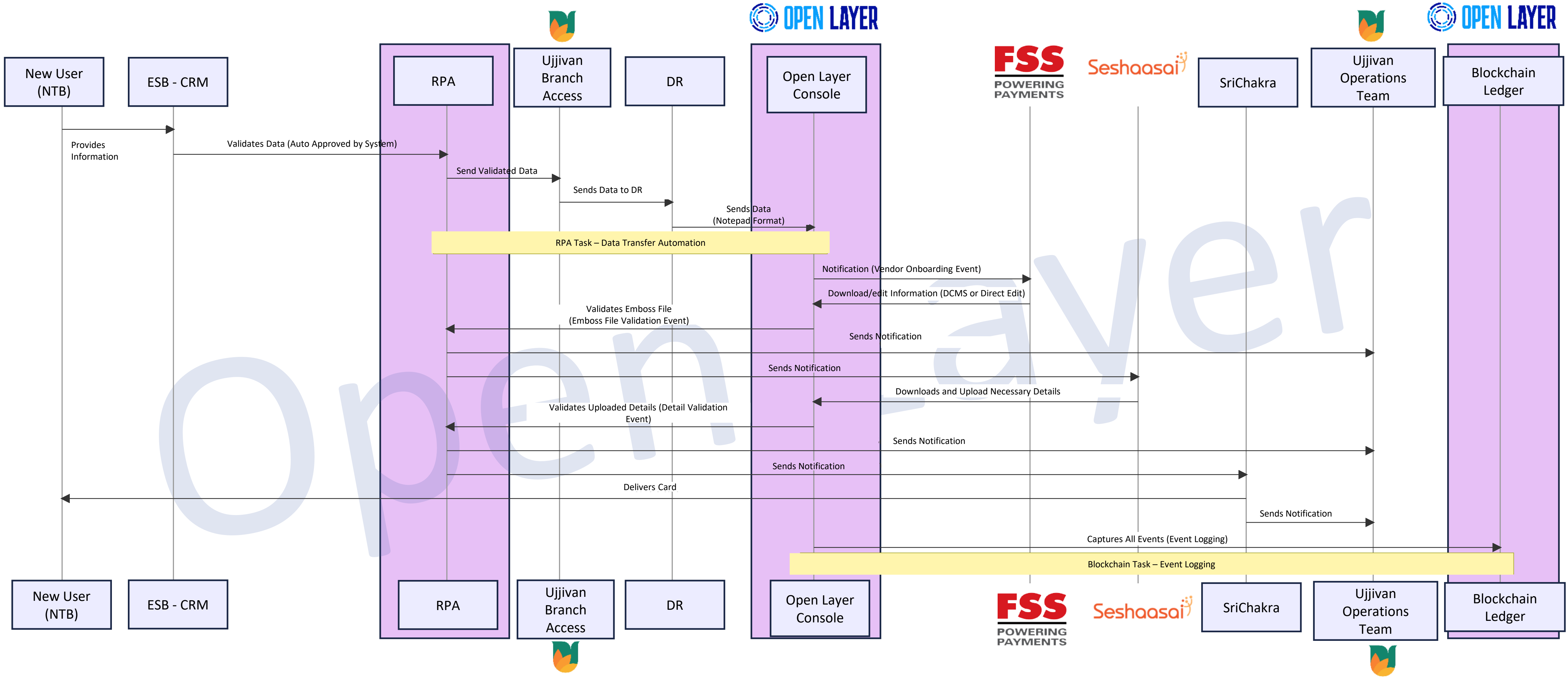
# Manual Interventions in Current Process

| S.No | Flow | Actor | Manual Intervention | File Formats |
|---|---|---|---|---|
| 1 | NTB | Branch/Sales Team | Raising personalized debit card requests through CRM/HHD | - |
| 2 | NTB | CRM | DRs will be verified, Auto-DR will be approved in the CRM | - |
| 3 | NTB | FU Officer | Downloading production file from network path and validating it | Notepad |
| 4 | NTB | FU Officer | Sending the validated printing file to FSS Vendor | SFTP with password protection |
| 5 | ETB | Customer | Applying for debit card re-issuance through various channels (branch, IBMB, phone banking) | - |
| 6 | ETB | Branch Manager/BOO | Verifying DRs in CRM to move the request to “sent to vendor” bucket | - |
| 7 | ETB | FU Officer | Downloading production file from network path and validating it | Notepad |
| 8 | ETB | FU Officer | Sending the validated printing file to FSS Vendor | SFTP with password protection |
| 9 | Vendor Process | Vendor A-FSS | Downloading the debit card file from EFTP Solution | EFTP Solution |
| 10 | Vendor Process | Vendor A-FSS | Creating the embossing file and sending it to Vendor B-Seshasaai | - |
| 11 | Vendor Process | Vendor B-Seshasaai | Completing card production and sending embossed cards to Vendor C-Srichakra | - |
| 12 | Vendor Process | Vendor B-Seshasaai | Saving dispatch status details in central location | EFTP Solution |
| 13 | Dispatch Status Update in CRM | FU Officer | Downloading the dispatch status file and creating a new file for CRM upload | Excel converted to Txt |
| 14 | Dispatch Status Update in CRM | FU Officer | Uploading the dispatch status file/insertion file in CRM | Txt |

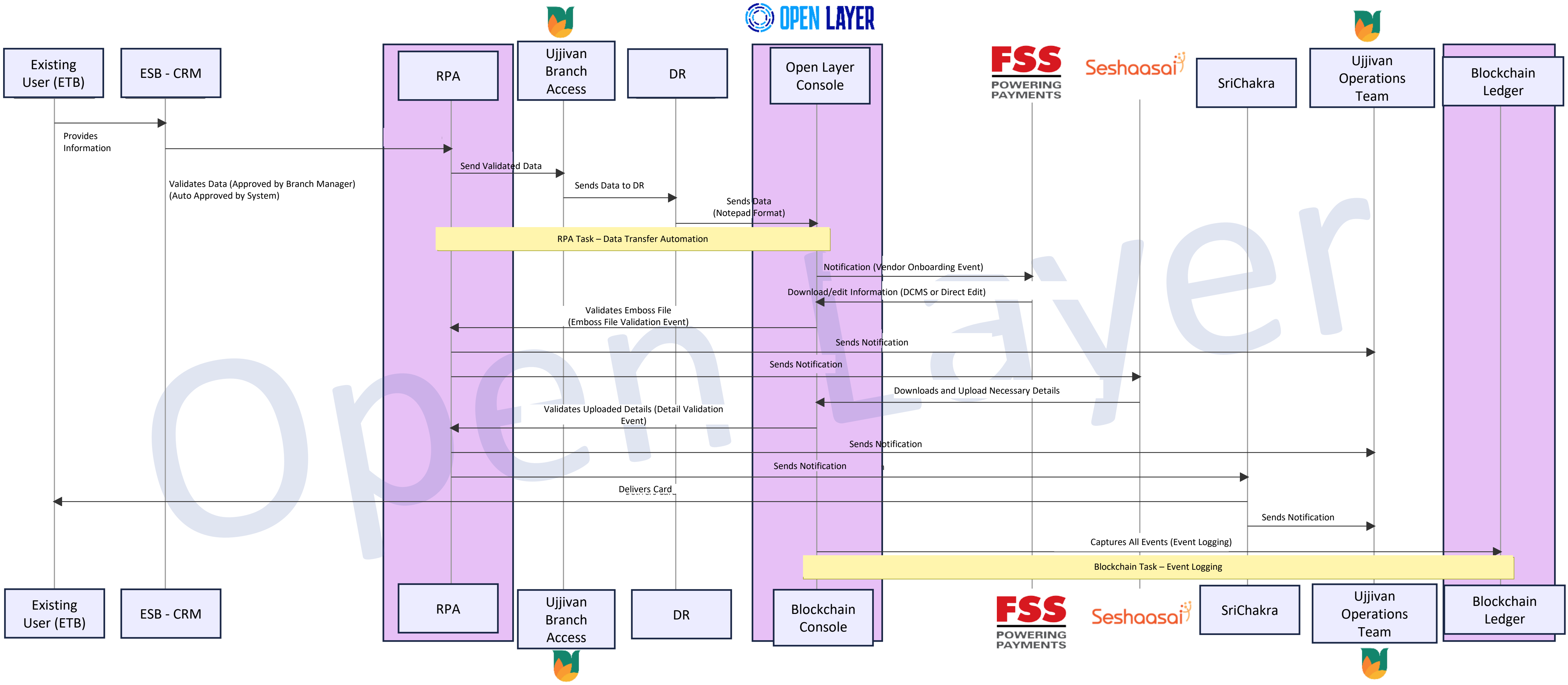
# Proposed High-Level Process



# Proposed Detailed Process (NTB)



# Proposed Detailed Process (ETB)

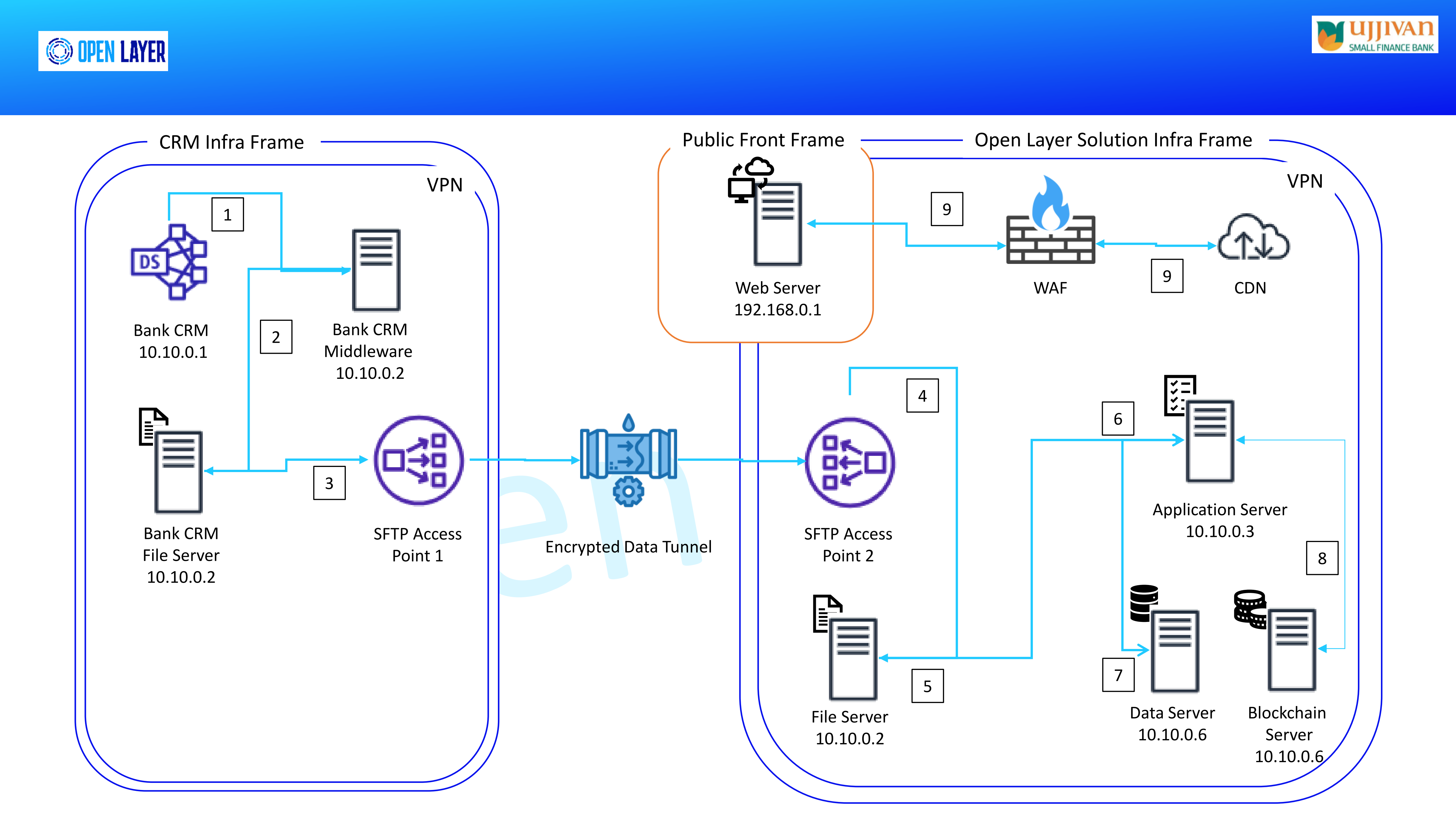


# Return of Investment (ROI) Aspects

| Benefit Category | Current Process | Proposed Process with RPA and Blockchain | Improvement Stats |
|---|---|---|---|
| **Operations Cost** | High due to manual interventions, multiple handoffs, and potential errors. | Significantly reduced due to automation, streamlined processes, and reduced errors. | **Estimated reduction of 60%** |
| **Manual FT Cost** | High costs associated with manual data entry, validation, and communication between teams. | Drastically reduced as RPA handles data validation, file transfers, and notifications. | **Estimated reduction of 70%** |
| **TAT - Turn Around Time (Debit Card Issuance)** | Longer due to manual processes, waiting times, and potential errors causing delays. | Faster due to automated validations, instant notifications, and streamlined vendor interactions. | **Estimated time reduction of 50%** |
| **Customer Experience and Retention (LTV)** | Potential dissatisfaction due to longer wait times and errors. Current LTV: 200 | Improved satisfaction due to faster service and fewer errors. Predicted increase in LTV due to better service. | **Estimated LTV increase to 440**<br>**120% increase** |
| **Customer Acquisition Cost (CAC)** | Higher costs due to delays in the user onboarding process and ease of user acquisition. | Reduced CAC as improved processes lead to positive customer feedback and referrals. | **Estimated CAC reduction by 80%** |

# Automation (Changes)

| Actor | Current Role & Responsibilities | Impact of Proposed Process | Changes in Role & Responsibilities |
|---|---|---|---|
| **Open Layer Console Admin** | - Onboard vendors manually.<br>- Communicate credentials via Email/SMS. | Centralized control and automation. | - Automated vendor onboarding.<br>- Automated credential dispatch via Email/SMS.<br>- Monitor blockchain logs. |
| **CRM System** | - Capture customer data.<br>- Manual validation.<br>- Send data to Branch. | Streamlined data flow and validation. | - Capture customer data.<br>- Automated validation via RPA.<br>- Direct communication with Console. |
| **Branch Manager/BOO** | - Verify and approve DRs<br>- Manual handoff to production. | Reduced manual verification. | - Oversee automated verifications.<br>- Monitor notifications and alerts. |
| **FU Officer** | - Download production file.<br>- Validate and send file manually to vendors. | Reduced manual tasks and errors. | - Monitor automated file transfers.<br>- Oversee RPA validations. |
| **Vendor A (FSS)** | - Manually download files;<br>- Create an embossing file.<br>- Send to Vendor B. | Streamlined data access and transfer. | - Access files via Open Layer Console.<br>- Automated notifications.<br>- Direct file transfer to Vendor B. |
| **Vendor B (Seshasaai)** | - Receive and process embossing file.<br>- Manually send cards to Vendor C. | Automated file reception and dispatch. | - Process embossing file.<br>- Automated dispatch to Vendor C via Open Layer Console. |
| **Vendor C (Srichakra)** | - Print and dispatch cards.<br>- Manual notifications to Ujjivan Bank. | Streamlined dispatch and notifications. | - Print and dispatch cards.<br>- Automated notifications via Console. |
| **Operations Team** | - Manual monitoring of process flow.<br>- Addressing issues and errors. | Reduced manual monitoring. | - Oversee automated processes.<br>- Address only major issues or escalations. |
| **Customer** | - Wait for longer processing times.<br>- Potential dissatisfaction due to errors. | Improved experience and faster service. | - Receive quicker services.<br>- Monitor status via notifications. |



OPEN LAYER
ujjivan SMALL FINANCE BANK
CRM Infra Frame
VPN
1
DS
Bank CRM 10.10.0.1
2
Bank CRM Middleware 10.10.0.2
Bank CRM File Server 10.10.0.2
3
SFTP Access Point 1
Encrypted Data Tunnel
Public Front Frame
Web Server 192.168.0.1
Open Layer Solution Infra Frame
VPN
9
WAF
9
CDN
4
SFTP Access Point 2
6
Application Server 10.10.0.3
8
File Server 10.10.0.2
5
7
Data Server 10.10.0.6
Blockchain Server 10.10.0.6

# Process & Data Flow

1. Customer information, along with debit card requests, is checked and processed.
2. The debit card generation file gets created and saved in the file server in an encrypted format with the preferred name convention.
3. The saved file now gets pushed through the SFTP access point using the process automation/batch process.
4. The file gets transferred through the secured encrypted data tunnel to the proposed Open Layer SFTP access point

1. The file is uploaded to the file server connected with the SFTP access point
2. The file is analyzed and processed in the standard format as prescribed / readable by DCMS [FSS] using process automation and generative AI
3. The processed information gets stored in RDMS along with the events index
4. The event logs that comprise of user activities, time-stamp, & stakeholder details gets captured in the Blockchain with reference to the events id
5. Application data is pulled & visualized in the unified platform with defined Role-based Access Controls for respective vendors and internal stakeholders

# Software Requirement Specifications

## Hardware Specifications

1. Web Server:
    1. Memory: 4 GB
    2. CPU: 2 Core
    3. IOPS: 300 – 500
    4. Storage: 50 GB
2. File Server (Can be reduced with compression):
    1. Storage: 500 GB
    2. IOPS: 500 – 750
    3. CPU: 4 Core
    4. Memory: 8 GB
3. App Server:
    1. Storage: 75 GB
    2. IOPS: 500 – 750
    3. CPU: 4 Core
    4. Memory: 8 GB

1. Database Server
    1. Storage: 75 GB
    2. IOPS: 500 – 750
    3. CPU: 4 Core
    4. Memory: 8 GB
2. Blockchain Server
    1. Storage: 750 GB
    2. IOPS: 750 – 1000
    3. CPU: 4 Core
    4. Memory: 8 GB

## Software Specifications

1. Preferred Operating System: Redhat 8, Ubuntu 20
2. Middleware Server: Jboss / Wildfly
3. Web Server: Apache 2 / Nginx
4. Database: Oracle 19g
5. Middleware Language : Java 11 / Go Lang
6. Frontend Framework: React JS
7. API MQ: Kafka / Active MQ / Reddis
8. MemCache: Reddis / ELK

**Note: DR can be set up with a similar config with respect to the App Server and Database. Firewalls and other security components are taken into consideration with the existing banking infrastructure**

# Get In Touch

- Open Layer specializes in providing banking solutions tailored to the specific needs of companies in the financial industry

- By leveraging our expertise, companies can unlock the full potential of blockchain banking services to enhance customer engagement, embrace digital transformation, and stay ahead in this rapidly evolving digital landscape

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